

V18 MCU validation — executive summary

175 s MCU capture aligned to manually annotated HR and breath-by-breath RR

MCU ↔ PC replay	PASS	175 rows and all decisions match; maximum HR numeric delta is 0.000008 bpm
RR vs 10 s trailing ref	PASS (this capture)	MAE 0.52 bpm; bias +0.003; 100% within 2 bpm
HR vs instantaneous ref	CONDITIONAL	MAE 3.43 bpm; bias +1.45; max error 12.67
HR vs 18 s trailing ref	CONDITIONAL	MAE 2.39 bpm; max error 9.01
Steady-state reporting	PASS	Continuous HR/RR report from 31–175 s; 31 s startup from recording start
Canine profile implementation	BLOCKED	Human profile is active; VITAL_MAX_BINS=64 truncates a naive PET 6–90 / 80–260 bpm scan
Canine evidence	NOT TESTED	No dog, coat, collar-placement, panting, posture-change or canine ECG/respiration ground truth

Conclusion

The STM32 migration is verified. The algorithm is suitable for controlled canine data collection, but it is not ready to provide trusted canine vital signs or production alerts.







